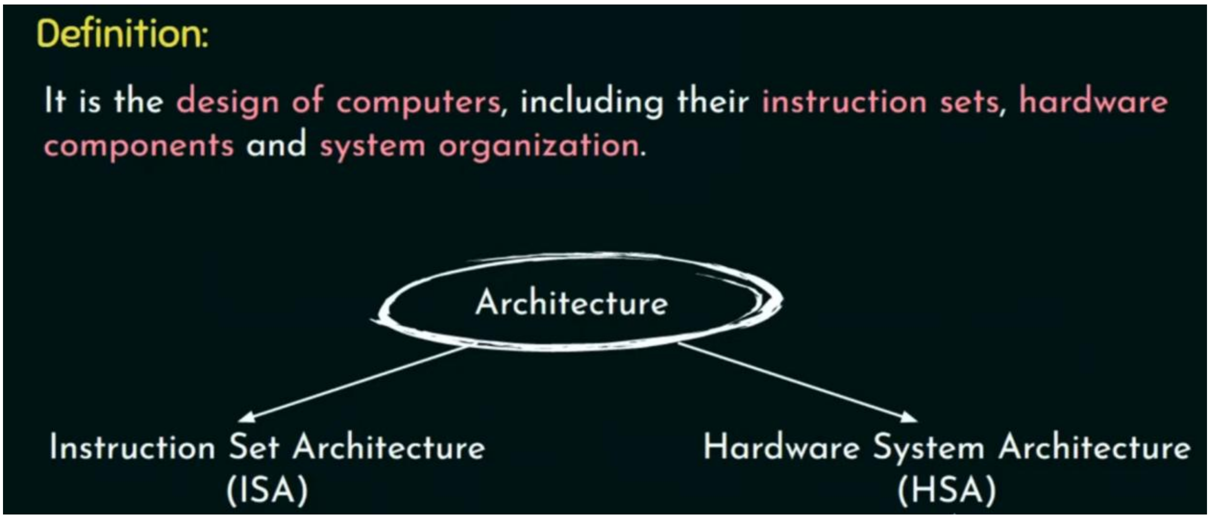
Computer Architecture

Basics of Computer Architecture



ISA: It includes the specifications that determine how machine language programs will interact with the computer.

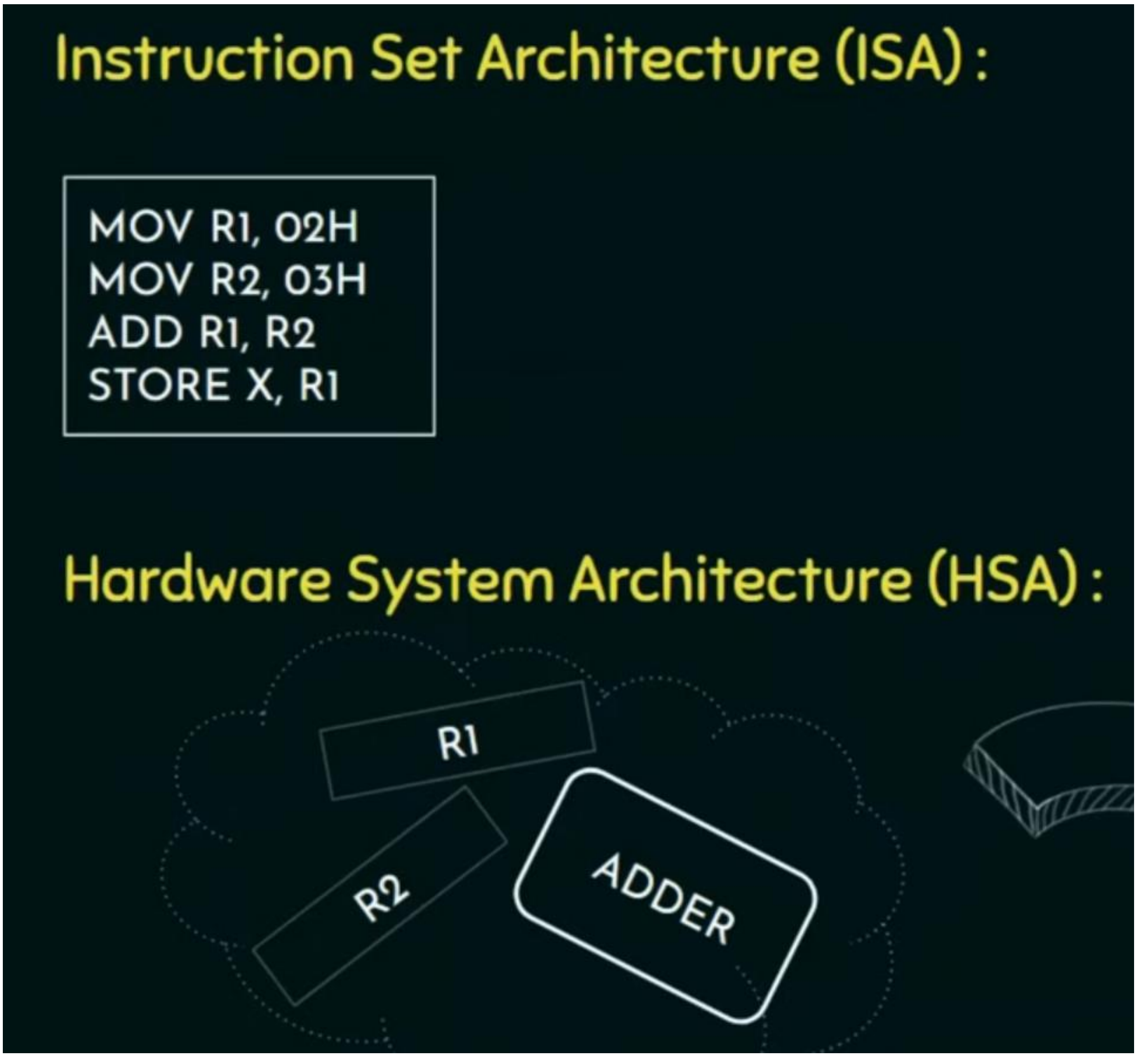
HSA: It deals with the computer's major hardware subsystems like CPU, storage, I/O, etc. It includes both the logical design and the data flow organization of the subsystems and hence determine the efficiency.

To solve x = 2 + 3 in computer:

We can have multiple type of computer based on hardware like:

ISA select one or more of available hardware. Suppose 2nd set of instructions are chosen then it will use similar set of instructions for other operations as well.

For this, we will need at least 2 different memory locations (Registers) and an Adder.

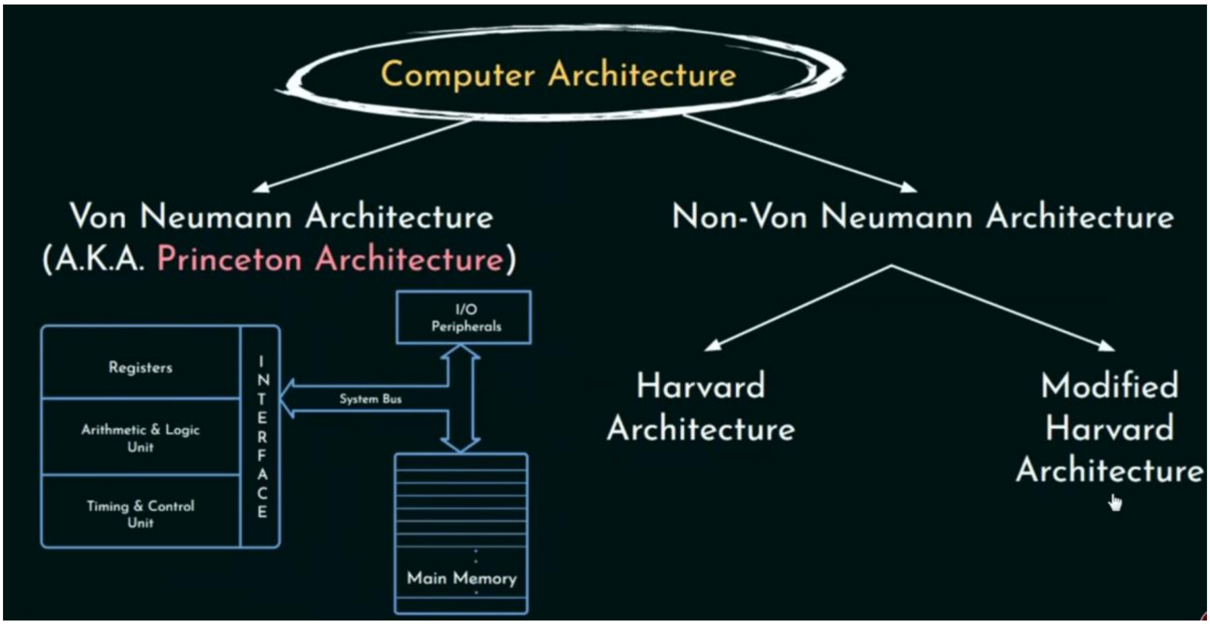


Therefore, we can say,

Implementation is the realization of computer in hardware and includes the choice of technology, speed, cost, etc.

But architecture does not define an implementation yet they both influence one another.

Classifications of Computer Architecture:

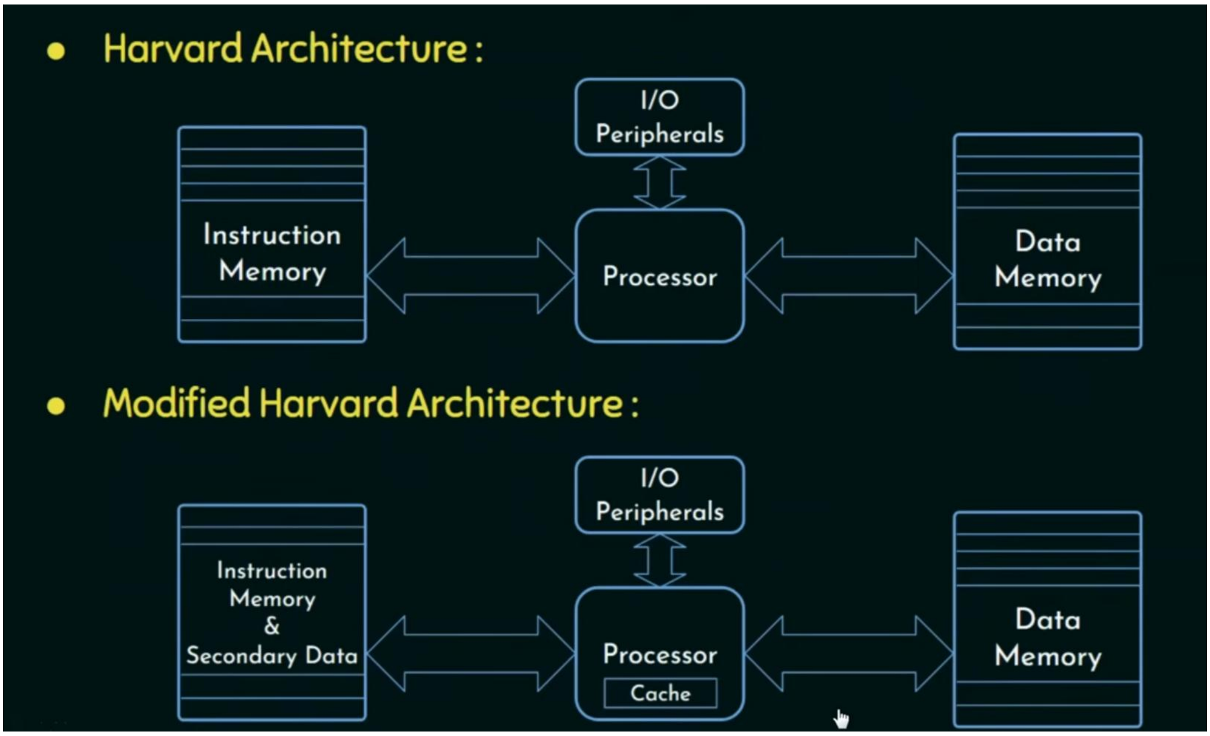


Von Neumann Architecture:

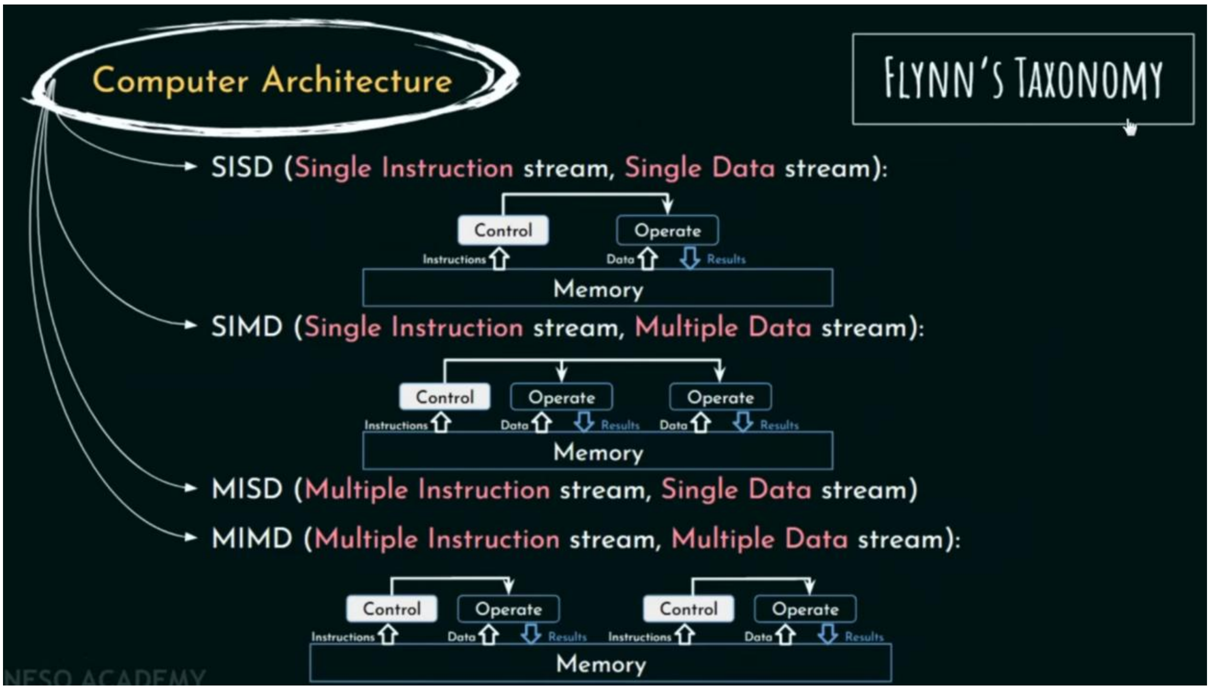
- It must be a stored program computer that means main memory (MM) holds the program which controls the computer operations and the computer can manipulate its own program more or less also it can do the same to any other data stored inside the memory furthermore it carries out instructions sequentially so CPU appears to execute one operation at a time.
- Major issue is having a single path between the main memory (MM) and the processor proposed by this architecture.
- **Bottleneck**:
  - It stores the instructions and the data in the same memory unit, thus both use same path means processor cannot simultaneously read an instruction and operate on data.

Non-Von Neumann Architecture:

- In Harvard, all from previous but they use 2 different memory units and thus pioneer parallelism this is popularly known as Harvard Architecture.

- 

CA are classified by a variety of characteristics including number of processors, number of programs they can execute, and the memory structure being used.



- SISD:
  - It has one CPU and executes one instruction at a time hence Single Instruction Stream.
  - Fetches and store one item of data at a time hence Single data Stream.
  - Von Neumann belongs to this category.

- SIMD:
  - It has multiple ALUs to operate but single hence Multiple Data Stream.
  - Processor array falls in this category
- MISD: Does not have any specific case but can be built using MIMD if needed.